GNU/Linux has a command that solves this problem. This is the 'rename' command, which renames the file names supplied according to the rule specified as the first argument. Let us understand this with a simple example.

Shown below are the files whose names contain spaces:

```
[bash]$ ls -1
song 1.mp3
song 2.mp3
song 3.mp3
song 4.mp3
```

Now, let us remove these spaces using the 'rename' command as follows:

```
[bash]$ rename 's/ //g' *.mp3
```

It's that simple; we are done. Now, let us verify the results:

```
[bash]$ ls -1
song1.mp3
song2.mp3
song3.mp3
song4.mp3
```

Indeed, it has worked. In the above example, the 'rename' command specified the rule to replace the space with nothing, i.e., it just removes the space. In the above command, the 'g' option implies 'global', i.e., 'remove *all* spaces'. If you want to replace spaces with any other character—for instance, an underscore—then use the following rule with the 'rename' command:

```
[bash]$ rename 's/ /-/g' *

[bash]$ ls -1
song-1.mp3
song-2.mp3
song-3.mp3
```

—Narendra Kangralkar, narendrakangralkar@gmail.com

Use *Tree* for visualising the directory tree structure

The *Tree* command is useful for visualising the directories and files residing in the current directory. To see what this command does, we need to install it first. Run the following command if you are on an Ubuntu based system:

```
sudo apt-get install tree
```

In Fedora-like systems, use the following command:

```
yum install tree
```

After installation, just type 'tree' and hit *Enter*. It will display the current directory tree structure with the file name in the respective directory.

If you just want to explore directories inside the current directory, type the following command:

```
tree -d
```

This will display the tree structure only with the listed directories. Other *tree* commands, and what they do, are listed below.

tree –a: Displays all files and directories

tree –h: Displays files and directories with their sizes

tree –f: Will display the tree structure with the full path to the file and directories

—Rajatkumar Zala, rajatkumarzala@yahoo.com

How to find out how long the process has been running

Here is a simple tip to get the PID of your Apache process. Type:

```
[root@linuxraja ~]# ps -ef | grep httpd | grep root | grep
-v grep

root     31096     1  0  2015 ?        00:00:52 /usr/sbin/
httpd
```

Now, we will check how long the process has been running:

```
ps -eo pid,etime  | grep PID

[root@linuxraja ~]# ps -eo pid,etime  | grep 31096

31096 152-22:29:16
```

The above output shows that the Apache process has been running for 152 days, 22 hours, 29 minutes and 16 seconds

—Natraj Solai, linuxraja@gmail.com

END

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The joy of using Linux is in finding ways to get around problems—take them head on, defeat them! We invite you to share your tips and tricks with us for publication in *OSFY* so that they can reach a wider audience. Your tips could be related to administration, programming, troubleshooting or general tweaking. Submit them at *www.opensourceforu.com*. The sender of each published tip will get a T-shirt.